

Computational Modeling in Python: Distributional Semantic Models



EMLaR 2025
Utrecht University
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Plan for today

1. Computational Modelling: what is it, and why should we care?
2. Our case study for today:
Learning Words from Context
3. Representing word meaning from context:
Distributional Semantic Models
4. Practical: let's implement the model!

Getting set up

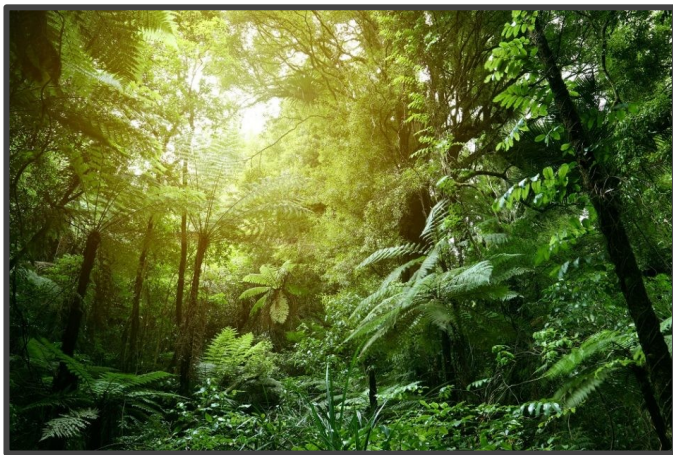
1. Install the [Anaconda](#) distribution of Python
2. Download the [tutorial materials](#) from GitHub
3. Open the Jupyter Notebook called [EMLAR2025.ipynb](#)

What is a model?

- A **model** is a **formal** and **simplified** representation of some aspect of reality

What is a Model?

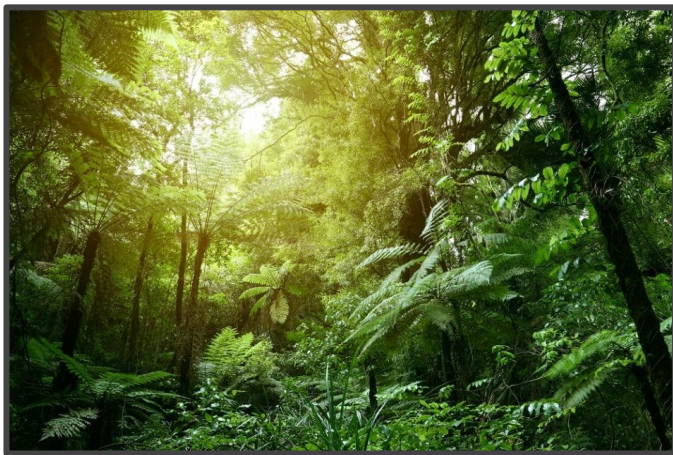
- A **model** is a **formal** and **simplified** representation of some aspect of reality



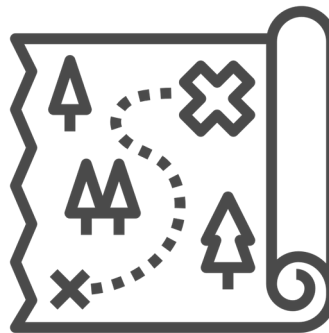
Reality

What is a model?

- A **model** is a **formal** and **simplified** representation of some aspect of reality



Reality



Simplification

What is a **computational** model?

- A **model** that can be described in a programming language as a sequence of steps that transform an input x into an output y .



Why should we use computational models?

- Formalization of a theory
 - Eliminate vagueness of verbal descriptions
- Prediction of unseen cases
- Explore causal explanations and build new theories (bottom-up)
- Automatization: we can apply (multiple) models to large amounts of data



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 - How do children segment words? How do people represent semantics?

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- 1000s of different architectures available
- Most models are designed to answer **specific questions**
 - How do children segment words? How do people represent semantics?
- But some are **unified theories** of cognition (e.g., Soar, ACT-R)
- There are some common ways of classifying cognitive models
 - Connectionist (Neural, Deep Learning, PDP), Symbolic, Bayesian

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- Often there are multiple ways to model the same behaviour or cognitive processes
- But this is not necessarily a problem
- It is often a good thing! Creates opportunities for comparing different theories

What kinds of computational models exist?

- Models can target different **levels of abstraction** (Marr, 1982)
 - What is learnable? How is it learned? How does this arise from neural structures?
- Different models might serve different goals, while still being derived from the same theory/hypotheses

All models are wrong but some are useful

- For language acquisition research, our main goal with modelling is to test causal hypotheses and make predictions
- Your model doesn't have to be an airtight explanation of every aspect of a phenomenon (semantics is very complex!)
- If the model provides a way of formally testing your ideas or can make testable predictions, then it is useful



Today: Learning Words From Linguistic Context

Word Learning

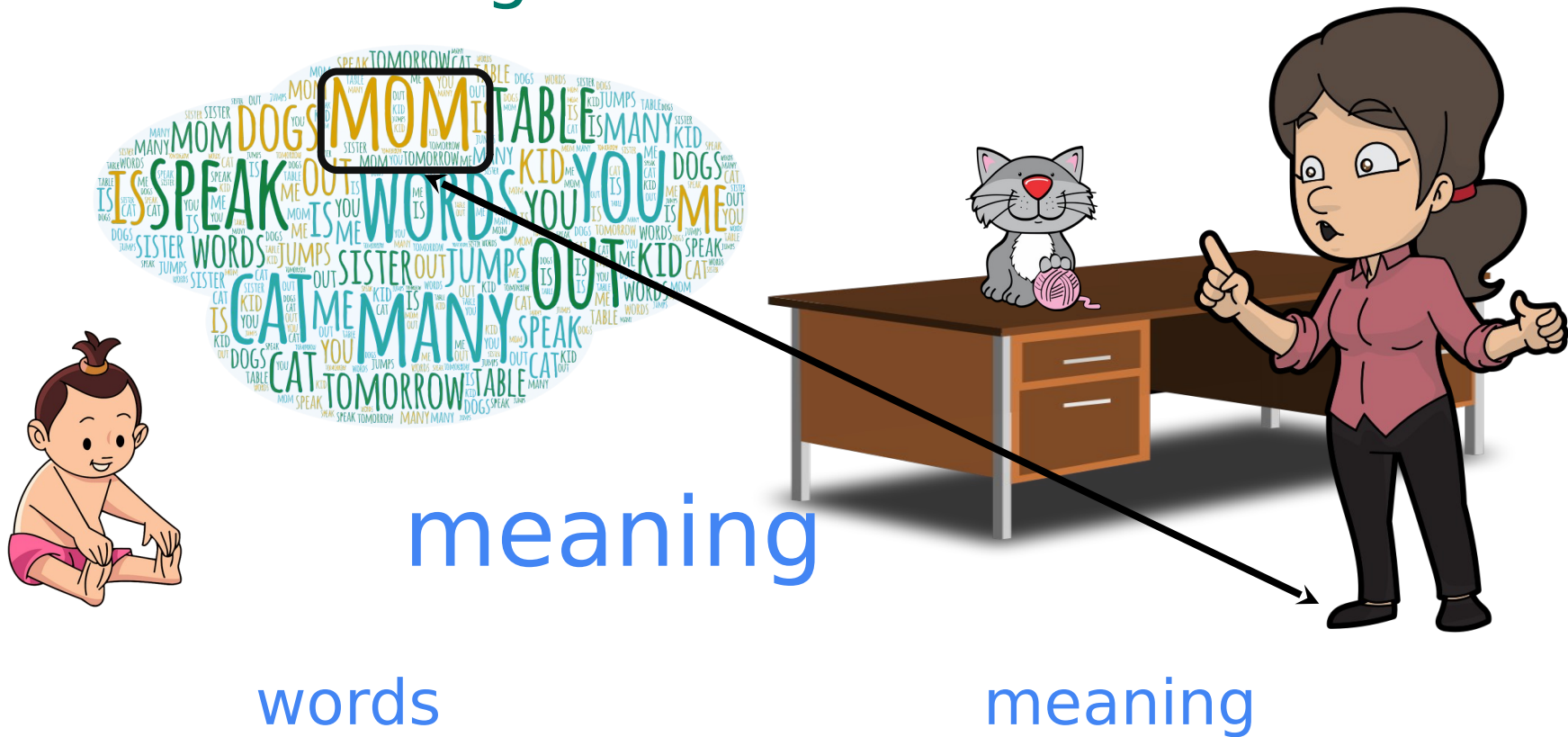


words



meaning

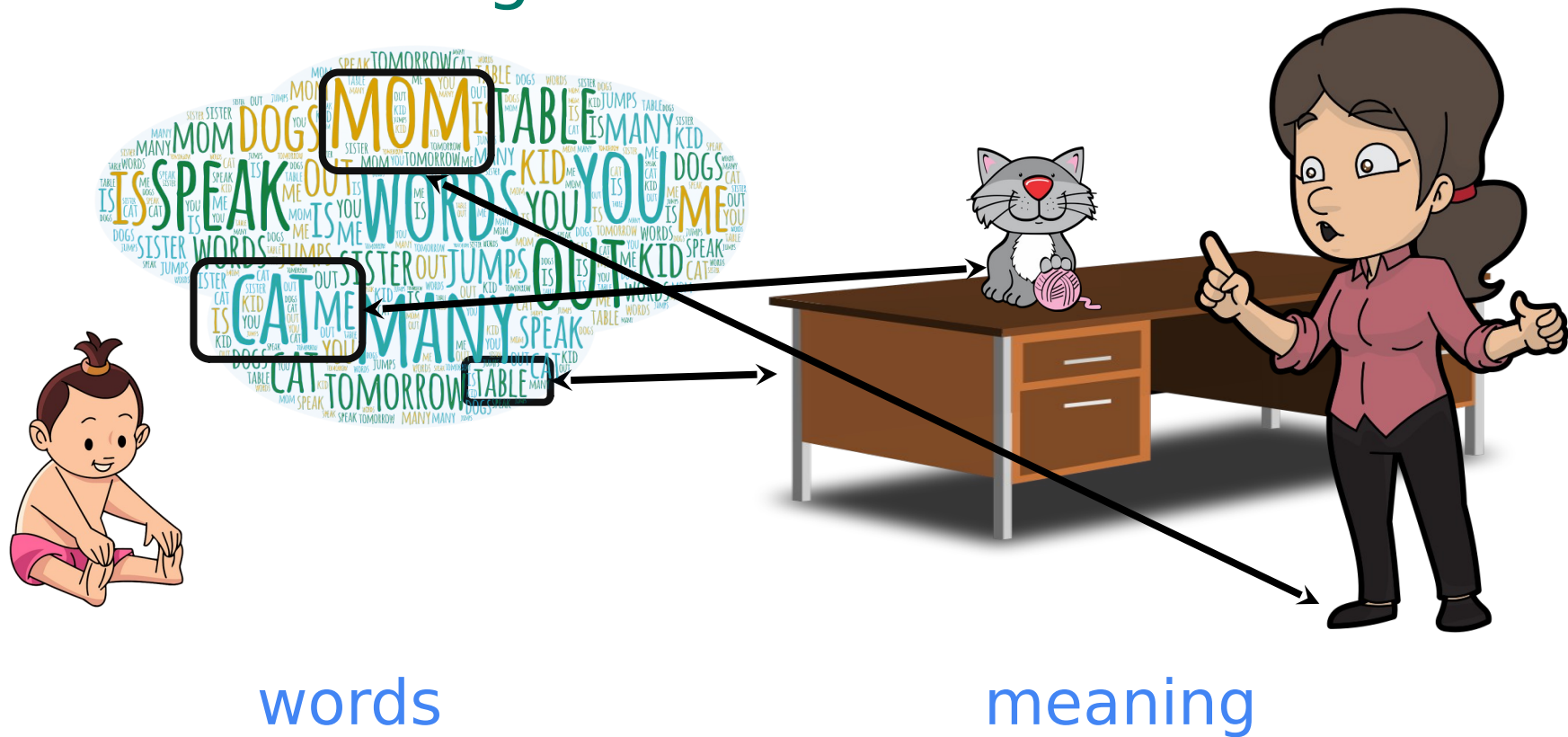
Word Learning



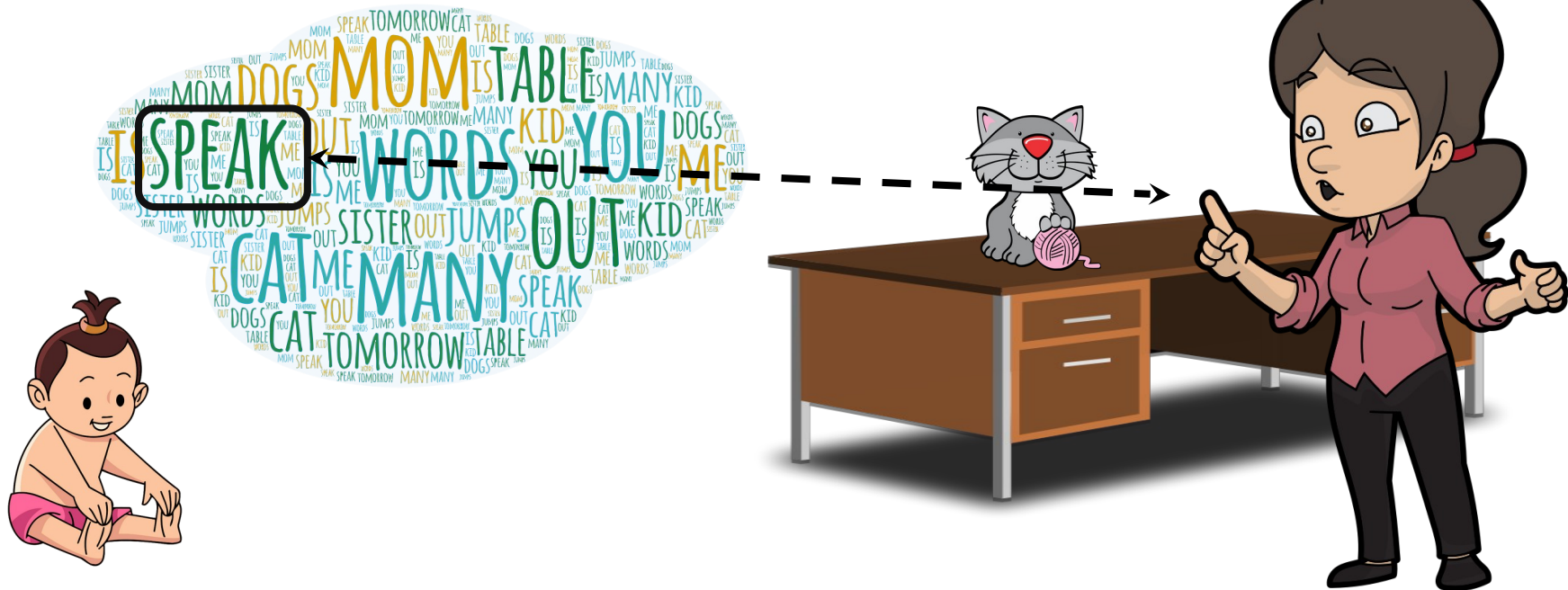
A cartoon illustration of a woman with dark hair in a ponytail, wearing a pink shirt and black pants, pointing her right index finger towards a grey cat sitting on a brown desk. The cat is holding a ball of pink yarn. In the background, a baby is sitting on the floor, and a large thought bubble contains various words like 'MOM', 'Dogs', 'Sister', 'You', 'Me', 'Cat', 'Table', 'Jumps', 'Words', 'Tomorrow', 'Many', 'Speak', 'Out', 'Kid', 'Is', 'Tombrow', 'Manny', 'Sister', 'Jumps', 'Table', 'Words', 'Tomorrow', 'Many', 'Speak', 'Out', 'Kid', 'Is', 'Tombrow', 'Manny'. Two black boxes highlight the words 'MOM' and 'ME' in the thought bubble, with arrows pointing from them to the woman and the cat respectively.

meaning

Word Learning



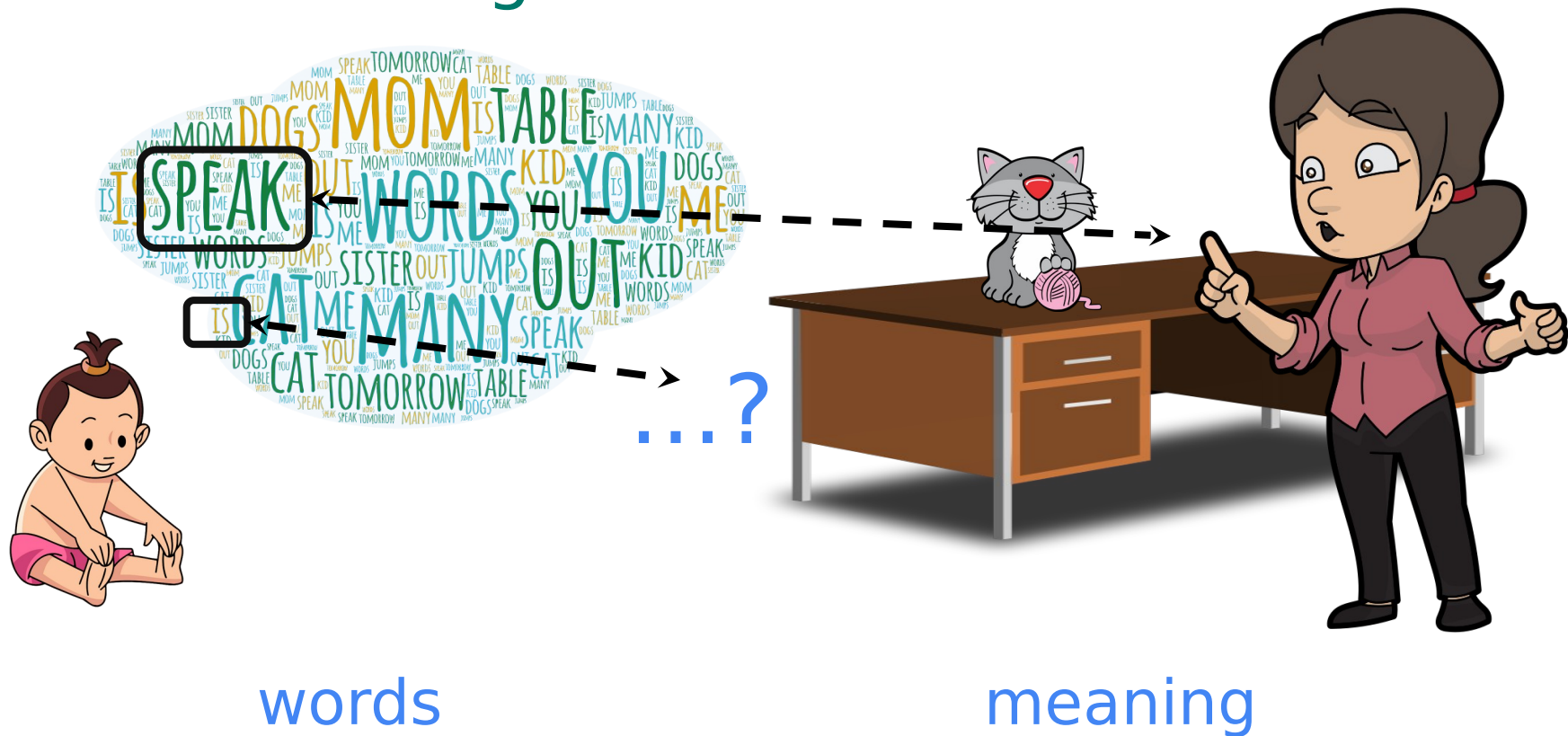
Word Learning



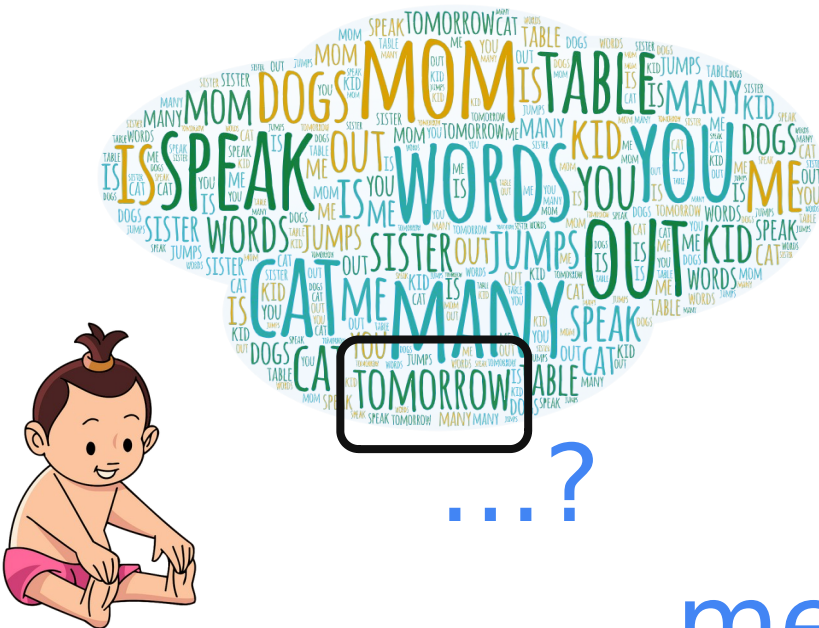
words

meaning

Word Learning



Word Learning



words

meaning

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Word Learning

- Mapping words to meaning is a difficult task
- How do children constrain the hypothesis space?



Word Learning

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- How do children constrain the hypothesis space?
 - Biases to select referent
 - mutual exclusivity bias, whole-object assumption, ...

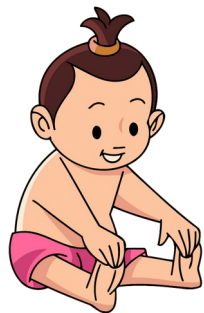


Word Learning

- Mapping words to meaning is a difficult task
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 - Cross-situational learning



But... words occur in continuous speech



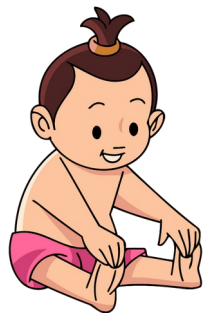
words



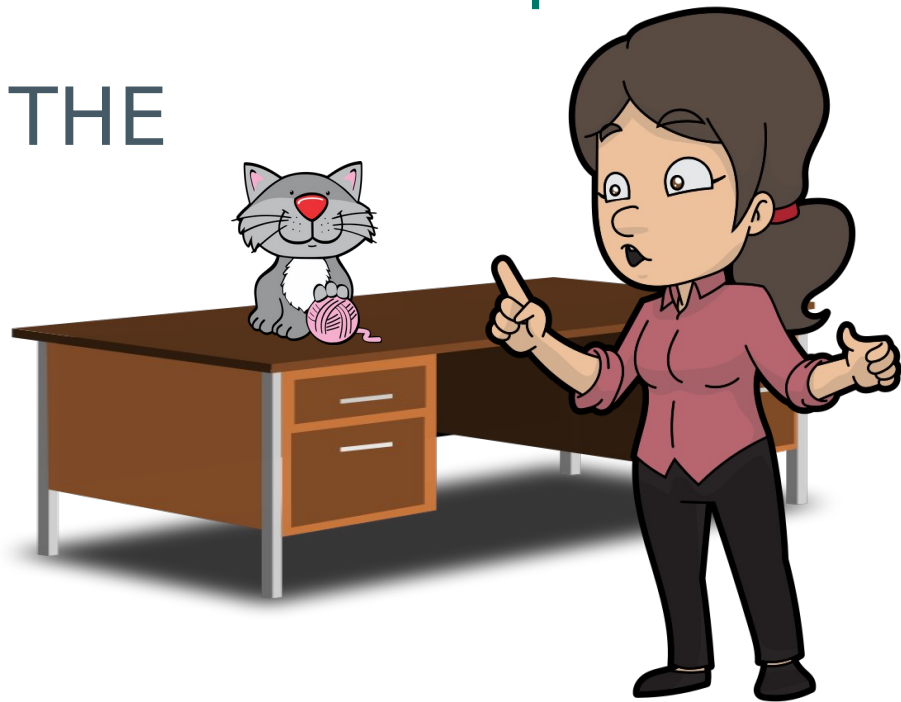
meaning

But... words occur in continuous speech

THE CAT IS ON THE
TABLE



words

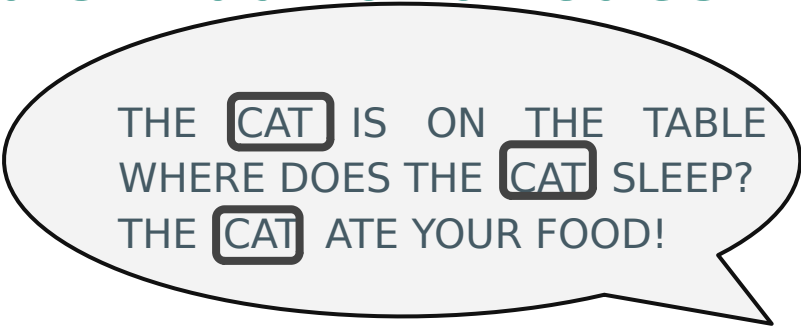


meaning

Word Learning

- Mapping words to meaning is a difficult task
- How do children constrain the hypothesis space?
 - Biases to select referent
 - mutual exclusivity bias, whole-object assumption, ...
 - Cross-situational learning
 - Distributional information from the linguistic input
 - Context

Word Learning from distributional cues



THE **CAT** IS ON THE TABLE
WHERE DOES THE **CAT** SLEEP?
THE **CAT** ATE YOUR FOOD!

- Children can track **distributional information** at a very young age (Saffran et al. 1996, Aslin et al. 1998, ...)
- How can distributional information shape meaning representations?
 - Let's look at **Distributional Semantic Models**

Distributional Semantic Models



Meaning as use

- “The meaning of a word is its use in the language”
(Wittgenstein, 1953)
- “What people know when they say that they know a word is not how to recite its dictionary definition – they know how to use it” (Miller, 1986)

The Distributional Hypothesis

- “Difference of meaning correlates with difference of distribution” (Harris, 1954)
- “You shall know a word by the company it keeps” (Firth, 1957)

Distributional Semantic Models

- **Distributional Semantic Models** (DSMs) derive semantic representations of words based on the distributional hypothesis
- These models are also known as **Vector Space Models**

Distributional Semantic Models

- The representations have the form of vectors (*embeddings*)

e.g.

249	0	365	446	0	0	90	0	2136
-----	---	-----	-----	---	---	----	---	------

- These representations are distributed rather than symbolic
 - similar words should have similar representations!

Distributional Semantic Models

- The number of words that we consider to be part of the context of another word is a parameter called "window size"
- We will now see an example of how we derive word representations in Distributional Semantic Models, for a model with window size = 2

Distributional Semantic Models

THE CAR TRAVELLED AT
HIGH SPEED

Distributional Semantic Models

THE CAR TRAVELLED AT HIGH SPEED

	the	car	travelled	at	high	speed	...
the	0	0	0	0	0	0	0
car	0	0	0	0	0	0	0
travelled	0	0	0	0	0	0	0
at	0	0	0	0	0	0	0
high	0	0	0	0	0	0	0
speed	0	0	0	0	0	0	0
...	0	0	0	0	0	0	0

Distributional Semantic Models

THE CAR TRAVELLED AT HIGH SPEED

	the	car	travelled	at	high	speed	...
the	0	1	1	0	0	0	0
car	0	0	0	0	0	0	0
travelled	0	0	0	0	0	0	0
at	0	0	0	0	0	0	0
high	0	0	0	0	0	0	0
speed	0	0	0	0	0	0	0
...	0	0	0	0	0	0	0

Distributional Semantic Models

THE **CAR** TRAVELLED AT HIGH SPEED

	the	car	travelle d	at	high	speed	...
the	0	1	1	0	0	0	0
car	1	0	1	1	0	0	0
travelle d	0	0	0	0	0	0	0
at	0	0	0	0	0	0	0
high	0	0	0	0	0	0	0
speed	0	0	0	0	0	0	0
...	0	0	0	0	0	0	0

Based on

Raquel G. Alhama & Andrew Jessop

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Distributional Semantic Models

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	the	car	travelled	at	high	speed	...
the	0	1	1	0	0	0	0
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travelled	1	1	0	1	1	0	0
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Distributional Semantic Models

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Distributional Semantic Models

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Distributional Semantic Models

THE CAR TRAVELLED AT HIGH SPEED

	the	car	travelled	at	high	speed	...
the	0	1	1	0	0	0	0
car	1	0	1	1	0	0	0
travelled	1	1	0	1	1	0	0
at	0	1	1	0	1	1	0
high	0	0	1	1	0	1	0
speed	0	0	0	1	1	0	0
...	0	0	0	0	0	0	0

Distributional Semantic Models

	the	car	travelled	at	high	speed	...
the	0	1	1	0	0	0	0
car	1	0	1	1	0	0	0
travelled	1	1	0	1	1	0	0
at	0	1	1	0	1	1	0
high	0	0	1	1	0	1	0
speed	0	0	0	1	1	0	0
...	0	0	0	0	0	0	0

Distributional Semantic Models

THE TRUCK TRAVELLED FAST

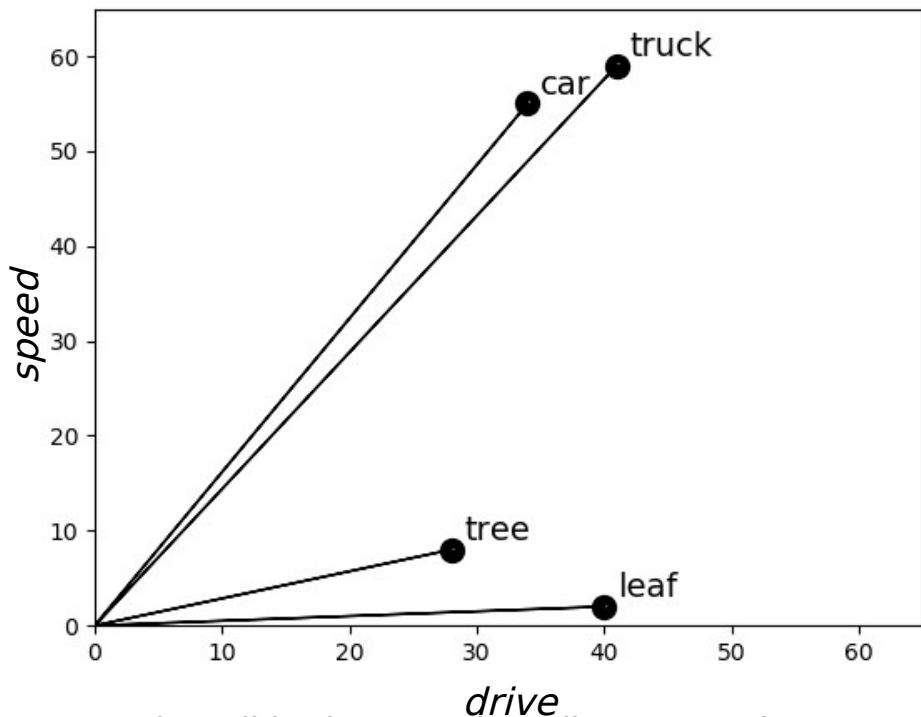
	the	car	travelled	at	high	speed	truck	fast
the	0	1	2	0	0	0	1	0
car	1	0	1	1	0	0	0	0
travelled	2	1	0	1	1	0	1	1
at	0	1	1	0	1	1	0	0
high	0	0	1	1	0	1	0	0
speed	0	0	0	1	1	0	0	0
truck	1	0	1	0	0	0	0	1
fast	0	0	1	0	0	0	1	0

Distributional Semantic Models

	the	car	travelled	at	high	speed	truck	fast
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high	0	0	1	1	0	1	0	0
speed	0	0	0	1	1	0	0	0
truck	1	0	1	0	0	0	0	1
fast	0	0	1	0	0	0	1	0

- This is a term-term *co-occurrence matrix* (for a toy example)
- The vectors reflect the use of words in real linguistic productions.
- Similar vectors for semantically related words!

Distributional Semantic Models



Based on slides by Raquel G. Alhama & Andrew Jessop

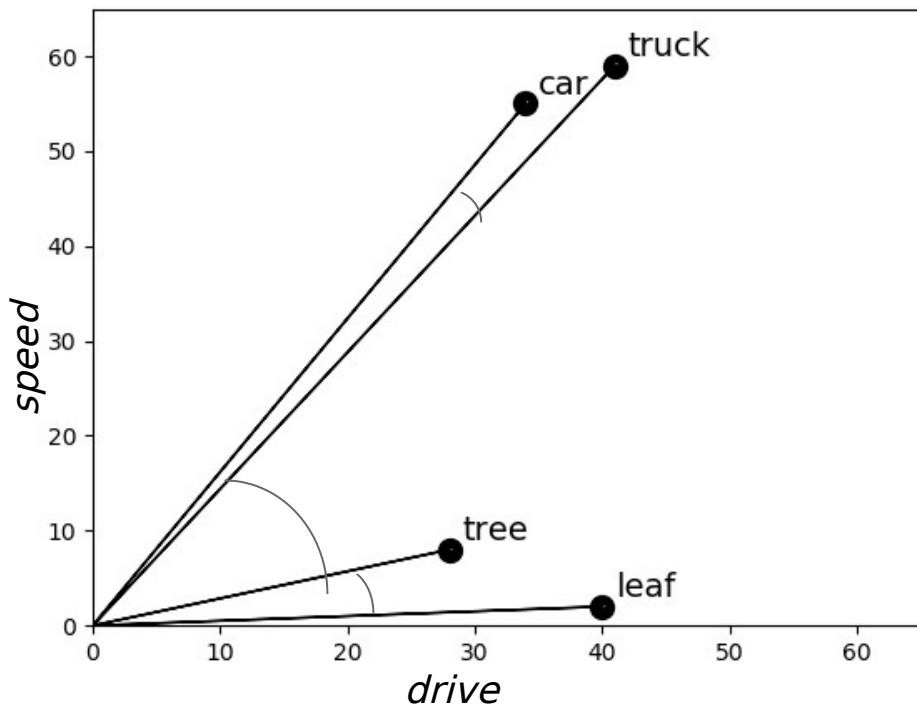
- We can interpret each position in an n -dimensional vector as a coordinate of a point in n -dimensional vector space

(graph reduced to 2 dimensions)
- Thus, these word representations have **geometric** properties (e.g. distance between words!)

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Distributional Semantic Models



- Cosine similarity: cosine of the angle of the vectors

$$\cos(\Theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

The cosine is between $[-1, 1]$.
A cosine similarity of 0 indicates orthogonal vectors (completely unrelated).

Let's implement this!

